Date :

Need the following values for rear - 1, rear - 2 spring on left & right side

Left

Rear -1 β_{2L} → Installation angle of rear -1 leaf spring

$$l_{i2} = l_2 c\alpha - \left(\cancel{\cos}\, \alpha\, l_2 + l_1 \sin\left(\cos^{-1}\left(\frac{V_1 - V_r}{l_1}\right)\right)\right) \cos\alpha$$

Need values l_2, l_1, $\begin{Bmatrix} V_1 \\ V_r \end{Bmatrix}$, α

$$(L_D + \Delta l) = (x_1 + l_2 + x_L)_{2L}$$

$\Rightarrow$ x_1, x_L, l_2, l_1, V_1, V_r, α

Rear-2 β_{3L} → Installation angle of rear-2 leaf spring

$$l_{i3} = L_3 c\alpha - \left(l_2 + l_1 \sin\left(\cos^{-1}\left(\frac{V_1 - V_r}{l_1}\right)\right)\right) \cos\alpha$$

$$L_D + \Delta l = (x_1 + l_2 + X_L)_{3L}$$

$\Rightarrow$ x_1, x_L, l_2, l_1, V_1, V_r, α